Partial Fractions by the Cover-Up Method

APSC 2113 – Engineering Analysis I
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Four partial-fraction problems, each solved with the cover-up method and checked.

1 The Cover-Up Method

Simple linear factor $(s-a)$.

1. Cover up (delete) that factor in the original fraction.
2. Substitute $s=a$ into what remains. The result is the coefficient over $(s-a)$.

Irreducible quadratic factor, e.g. (s^2+1).

1. Cover up (s^2+1) and substitute $s=i$, a root of $s^2+1=0$.
2. The result is a complex number $Bi+C$. Match real and imaginary parts to obtain the numerator $Bs+C$.

In the steps below, the covered-up factor is shown in a shaded box.

2 Problem 1: $\dfrac{1}{s(s+1)}$

Step 1: Set up the form.

$$\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1}$$

Step 2: Find A. Cover up s and set $s=0$:

$$A = \left.\frac{1}{\boxed{s}\,(s+1)}\right|_{s=0} = \frac{1}{0+1} = 1$$

Step 3: Find B. Cover up $(s+1)$ and set $s=-1$:

$$B = \left.\frac{1}{s\,\boxed{(s+1)}}\right|_{s=-1} = \frac{1}{-1} = -1$$

Answer:

$$\boxed{\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}}$$

Check: $\dfrac{(s+1)-s}{s(s+1)} = \dfrac{1}{s(s+1)}$ ✓

3 Problem 2: $\dfrac{5s}{(s+2)(s^2+1)}$

Step 1: Set up the form. The quadratic factor needs a linear numerator:

$$\frac{5s}{(s+2)(s^2+1)} = \frac{A}{s+2} + \frac{Bs+C}{s^2+1}$$

Step 2: Find A. Cover up $(s+2)$ and set $s=-2$:

$$A = \left.\frac{5s}{(s+2)\,(s^2+1)}\right|_{s=-2} = \frac{-10}{4+1} = -2$$

Step 3: Find B and C. Cover up (s^2+1) and set $s=i$:

$$Bi + C = \left.\frac{5s}{(s+2)\,(s^2+1)}\right|_{s=i} = \frac{5i}{2+i}$$

Rationalize by multiplying the top and bottom by the conjugate $(2-i)$:

$$\frac{5i(2-i)}{(2+i)(2-i)} = \frac{10i-5i^2}{4+1} = \frac{10i+5}{5} = 1+2i$$

Match parts: $Bi+C = 2i+1$, so $B=2$ and $C=1$.

Answer:

$$\boxed{\frac{5s}{(s+2)(s^2+1)} = \frac{-2}{s+2} + \frac{2s+1}{s^2+1}}$$

Check: The combined numerator is

$$-2(s^2+1) + (2s+1)(s+2) = -2s^2 - 2 + 2s^2 + 5s + 2 = 5s \;\checkmark$$

4 Problem 3: $\dfrac{10}{(s+1)(s+2)(s^2+1)}$

Step 1: Set up the form.

$$\frac{10}{(s+1)(s+2)(s^2+1)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{Cs+D}{s^2+1}$$

Step 2: Find A. Cover up $(s+1)$ and set $s=-1$:

$$A = \left.\frac{10}{(s+1)\,(s+2)(s^2+1)}\right|_{s=-1} = \frac{10}{(1)(2)} = 5$$

Step 3: Find B. Cover up $(s+2)$ and set $s=-2$:

$$B = \left.\frac{10}{(s+1)\,(s+2)\,(s^2+1)}\right|_{s=-2} = \frac{10}{(-1)(5)} = -2$$

Step 4: Find C and D. Cover up (s^2+1) and set $s=i$:

$$Ci + D = \left.\frac{10}{(s+1)(s+2)\,(s^2+1)}\right|_{s=i} = \frac{10}{(i+1)(i+2)}$$

Expand the denominator: $(i+1)(i+2) = i^2 + 3i + 2 = 1 + 3i$. Then rationalize:

$$\frac{10}{1+3i} \cdot \frac{1-3i}{1-3i} = \frac{10(1-3i)}{1+9} = 1 - 3i$$

Match parts: $Ci + D = -3i + 1$, so $C = -3$ and $D = 1$.

Answer:

$$\boxed{\frac{10}{(s+1)(s+2)(s^2+1)} = \frac{5}{s+1} - \frac{2}{s+2} + \frac{1-3s}{s^2+1}}$$

Check (plug in values):

- At $s=0$: original $= \frac{10}{2} = 5$; decomposition $= 5 - 1 + 1 = 5$ ✓
- At $s=1$: original $= \frac{10}{12} = \frac{5}{6}$; decomposition $= \frac{5}{2} - \frac{2}{3} - 1 = \frac{5}{6}$ ✓

5 Problem 4: $\dfrac{1}{(s+1)(s+2)}$

Step 1: Set up the form.

$$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

Step 2: Find A. Cover up $(s+1)$ and set $s=-1$:

$$A = \left.\frac{1}{(s+1)\,(s+2)}\right|_{s=-1} = \frac{1}{-1+2} = 1$$

Step 3: Find B. Cover up $(s+2)$ and set $s=-2$:

$$B = \left.\frac{1}{(s+1)\,(s+2)}\right|_{s=-2} = \frac{1}{-2+1} = -1$$

Answer:

$$\boxed{\frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}}$$

Check: $\dfrac{(s+2)-(s+1)}{(s+1)(s+2)} = \dfrac{1}{(s+1)(s+2)}$ ✓

6 Inverse Laplace Transforms

Using the standard pairs

$$\mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}, \qquad \mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} = \cos t, \qquad \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t,$$

each decomposition inverts term by term:

Problem	$f(t)$
1	$1 - e^{-t}$
2	$-2e^{-2t} + 2\cos t + \sin t$
3	$5e^{-t} - 2e^{-2t} - 3\cos t + \sin t$
4	$e^{-t} - e^{-2t}$